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Class 11 Chemistry Notes Chapter 11 The p-block elements

Introduction:

The elements in which last electron enters into p-subshell are called as p-block elements. The number of p-orbitals is three and, therefore, the maximum number of electrons that can be accommodated in a set of p-orbitals is six, hence p-block contains six groups.

Boron Family

Group III A contains six elements: Boron, aluminium, gallium, indium, thallium and ununtrium. The penultimate shell (next to the outermost) conains $1s^2$ in boron, $2s^2$ $2p^6$ (8

electrons) in aluminium and $(n-1)s^2(n-1)p^6(n-1)d^{10}$ (18 electrons) in other elements. Boron is a non-metal and always form covalent bonds. Boron family is known as most heterogeneous family as there is no regular trend in all properties, as it comes after d-block, lanthanoid contraction, poor shielding of d-orbital, they have large deviation in properties.

1. Physical Properties

The atomic radius, ionic radius and density increases when one moves from top to bottom in a group in periodic table. While melting point decreases from B to Ga and then increases from (Ga to In). Ionisation energy decreases from B to Al, but shows a reverse trend in going from Al to Ga.

2. Chemical Properties

i. Reaction with air: Impure boron in air forms oxide while pure boron is less reactive.

$$4B + 3O_2 \rightarrow 2B_2O_3$$

ii. Reaction with water: Boron is not affected by water or steam under ordinary conditions. However, Aluminium reacts with cold water if oxide layer is not present on its surface.

$$4Tl + 2H_2O + O_2 \rightarrow 4TlOH$$

iii. Reaction with acids: Boron is not affected by non-oxidising acids like HCl and dilute H_2SO_4 while other elements dissolve and liberate H_2 gas.

$$2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$$

iv. Reaction with alkalies: Boron, Aluminium, Gallium react with alkali solutions whereas Indium and Thallium are not affected by alkalies.

$$2B + 6NaOH \rightarrow 2Na2BO_3 + 3H_2$$

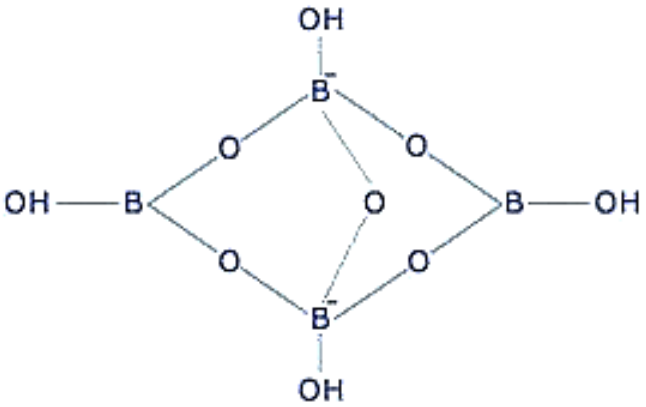
Anomalous Properties of Boron

Boron, the first member of group 13 elements, shows anomalous behaviour and differ from rest of the members of its family. The main reason for this difference are:

- Exceptionally small atomic and ionic size.
- High ionization enthalpy.
- Absence of d orbital in its valence shell.
- It has higher melting and boiling point than those of the other members of its group.

Borax

It Borax contains the tetranuclear unit i.e. $[B_4O_5(OH)_4]^{2-}$. The correct formula is $Na_2[B_4O_5(OH)_4].8H_2O$.



1) From Tincal

Naturally occurring borax is called tincal or suhaga. Tincal obtained from dried up lakes contains. 50% borax. It is boiled with water and filtered to remove insoluble impurities of clay,sand. The filtrate is concentrated when crystals of borax separate out.

2) From Colemanite

The mineral colemanite is finely powdered and is boiled with sodium carbonate solution.

$$Ca_2B_6O_{11} + 2\ Na_2CO_3 \rightarrow Na_2B_4O_7 + 2\ NaBO_2 + 2\ CaCO_3\downarrow$$

The precipitate of calcium carbonate thus formed is removed by filtration. The filtrate is concentrated and cooled when crystals of borax separate out. Sodium metaborate present in the mother liquor can be converted into borax by passing a current of carbon dioxide through it.

3) From Boric Acid

Borax can also be prepared in small amounts by neutralising boric acid with sodium carbonate.

On cooling crystal of borax i.e. $Na_2B_4O_7.10H_2O$ separate out.

Properties of Borax

1) It is a white crystalline solid, less soluble in cold water but more soluble in hot water.

2) The aqueous solution of borax is alkaline due to hydrolysis. Borax is therefore, used as a water softener and cleaning agent.

$$Na_2B_4O_7 + 2\ H_2O \rightarrow 2\ NaOH + H_2B_4O_7$$

$$Na_2B_4O_7 + 5\ H_2O \rightarrow 4H_3BO_3$$

Action of heat- borax Bead Test

Borax loses its water of crystallisation and swells up to form a puffy mass. On further heating, it melts into a clear liquid which solidifies to a transparent glass like bead which consists of sodium metaborate ($NaBO_2$) and boric anhydride (B_2O_3)

$$Na_2B_4O_7.10H_2O \rightarrow Na_2B_4O_7 + 10\ H_2O$$

$$Na_2B_4O_7 \rightarrow 2\ NaBO_2 + B_2O_3$$

The glassy bead is commonly known as borax bead and is employed in qualitative analysis for the detection of certain coloured basic radicals such as Ni^{2+}, Co^{2+}, Cr^{3+}, Cu^{2+}, Mn^{2+}. Whenever a coloured salt containing these cations is heated with borax bead on a platinum wire, the salt decomposes to form the corresponding metal oxide which then combines with B_2O_3 present in the glassy bead to form coloured metaborates. This test is called borax bead test.

$COSO_4 \rightarrow CoO + SO_3$

$CoO + B_2O_3 \rightarrow Co(BO_2)_2$

$NiO + B_2O_3 \rightarrow Ni(BO_2)_2$

$Cr_2O_3 + B_2O_3 \rightarrow 2\ Cr(BO_2)_3$

$MnO + B_2O_3 \rightarrow Mn(BO_2)_2$

$CuO + B_2O_3 \rightarrow Cu(BO_2)_2$

Action of Sodium Hydroxide

On adding a calculated quantity of sodium hydroxide to borax, sodium metaborate is formed.

$Na_2B_4O_7 + 2\ NaOH \rightarrow Na_2BO_2 + H_2O$

Action of Sulphuric Acid

On adding a calculated quantity of concentrated sulphuric acid to a hot concentrated solution of borax, boric acid is produced.

$Na_2B_4O_7 + H_2SO_4 \rightarrow Na_2SO_4 + H_2B_4O_7$

$H_2B_4O_7 + 5\ H_2O \rightarrow 4\ H_3BO_3$

Action of Ethyl Alcohol and Sulphuric Acid

On heating borax with ethyl alcohol and concentrated sulphuric acid vapours of triethylborates are produced. When ignited these vapours burn with a green edged flame.

$$Na_2B_4O_7 + H_2SO_4 + 5\ H_2O \rightarrow Na_2SO_4 + 4\ H_2BO_3$$

$$H_3BO_3 + 3\ C_2H_5OH \rightarrow B(OC_2H_5)_3 + 3\ H_2O$$

Uses of Borax

Borax is used in:

1) in the manufacture of enamels and glazes for earthen ware. The glazed surface is resistant to heat, stains and scratches.

2) as a flux in soldering

3) in the preparation of medicinal soaps due to its antiseptic properties.

4) in the manufacture of heat resistant Borosilicate glass.

5) to make peroxoborate- an important cleansing and bleaching agent present in washing powder.

6) as a stiffening agent for candle wicks.

7) to produce a good finish in laboratory.
8) in the laboratory for borax bead test.
9) in softening of water.

Boron Hydrides

The binary compounds of boron with hydrogen are called Boron hydrides.
These hydrides of boron are called boranes. These hydrides can be divided into several series:
1) B_nH_{n+4} (called nido borane) such as B_2H_6 (diborane), B_5H_9 (pentaborane), B_6H_{10} (hexaborane), B_8H_{12} (octaborane), $B_{10}H_{14}$ (decaborane).
2) B_nH_{n+6} (called arachno borane) such as B_4H_{10} (tetraborane), B_5H_9 (pentaborane), B_6H_{10} (hexaborane), B_8H_{14} (octaborane), B_9H_{15} (nonaborane).

Preparation

1) By reduction of boron trifluoride etherate with Lithium aluminium hydride in diethyl ether.

$$4\ BF_3.Et_2O + 3\ LiAlH_4 \rightarrow 2\ B_2H_6 + 3\ LiF + 3\ AlF_3 + 4\ Et_2O$$

2) Diborane is prepared in the laboratory by the oxidation of sodium borohydride with iodine in diglyme as solvent.

$$NaBH_4 + I_2 \rightarrow B_2H_6 + 6\ NaF$$

3) On the industrial scale, diborane is obtained by the reduction of BF_3 with sodium hydride.

$$BF_3 + 6\ NaH \rightarrow B_2H_6 + 6\ NaF$$

Preparation of higher boranes

Higher boranes are obtained when B_2H_6 is heated at 373-523 K.

Properties

1) Physical state

Diborane is a colourless, highly toxic gas, b.p. 180 K.

2) Stability

Diborane is stable only at low temperature. When diborane is heated in a sealed tube, between 373-533 K, a complex reaction occurs and various higher boranes are formed.

$$B_2H_6 \rightarrow B_4H_{10}, B_5H_{11}, B_6H_{12}$$

By careful control of temperature pressure and reaction time, different individual boranes can be obtained.

$$2\ B_2H_6 \rightarrow B_4H_{10} + H_2$$

3) Combustibility

It catches fire spontaneously upon exposure to air. It burns in oxygen evolving an enormous amount of heat.

$$B_2H_6 + 3O_2 \rightarrow B_2O_3 + 3\ H_2O\ ;\ A_cH° = -1976\ KJ/mol$$

Like diboranes, higher boranes spontaneously burn in air.

4) Hydrolysis

Boranes are readily hydrolysed by water to form boric acid.

$$B_2H_6 + 6\ H_2O \rightarrow 2\ H_3BO_3 + 6\ H_2$$

With methanol, trimethylborate is formed.

$$B_2H_6 + 6\ CH_3OH \rightarrow 2\ B\ (OCH_3)_3 + 6\ H_2$$

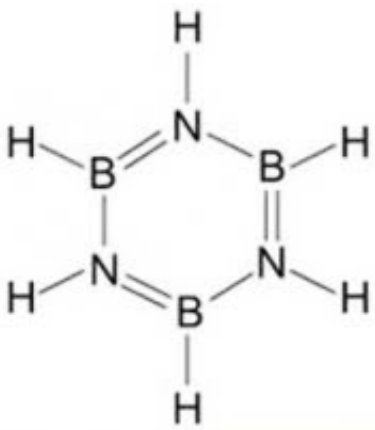
5) Reaction with Lewis bases – cleavage reaction

Diborane on treatment with Lewis bases first undergoes cleavage to form borane which then reacts with Lewis bases to form adducts.

$B_2H_6 + 2\ NMe_3 \rightarrow 2\ BH_3.NMe_3$

$B_2H_6 + 2\ CO \rightarrow 2\ BH_3.CO$

6) Reaction with Ammonia



Diborane combines with ammonia to form an addition product, $B_2H_6.2NH_3$ formulated as $[BH_2(NH_3)_2]^+[BH_4]^-$, which when heated to 473 K decompose to give a volatile compound called borazine or borazole.

$$3\ B_2H_6 + 6\ NH_3 \rightarrow 3\ [BH_2\ (NH_3)_2]^+\ [BH_4]^- \rightarrow 2\ B_3N_3H_6 + 12\ H_2$$

Borazine is isoelectronic and isosteric with benzene and its structure is similar to that of benzene expect that in benzene the π -electrons are completely delocalised but in borazine, they are only partially delocalized.

Because of its similarity with benzene, borazine is also called inorganic benzene.

7) Formation of complex borohydride

Several metal hydrides react with diborane to from tetrahydridoborates known as borohydride. All these contain the tetrahedral $[BH_4]^-$ ion.

$$2\ NaH + B_2H_6 \rightarrow 2\ Na^+\ [BH]^-$$

$$2LiH + B_2H_6 \rightarrow 2\ Li^+\ [BH_4]^-$$

Both sodium borohydride and lithium borohydride are used as reducing agent in organic synthesis. They also serve as starting material for many other borohydrides.

8) Reaction with alkalies

Diborane dissolves in strong alkali to produce metaborates and dihydrogen gas.

$$B_2H_6 + 2\ KOH + 2\ H_2O \rightarrow 2\ KBO_2 + 6\ H_2$$

9) Action of halogen acids

Diborane reacts with halogen acids to give halodiboranes evolving H_2 gas.

The reactivity of halogen acids follow the order: HI> HBr > HCl

Thus HI react at about 323 K in absence of catalyst while HBr and HCl reacts in presence of their aluminium halide as catalyst.

$$B_2H_6 + HI \rightarrow B_2H_5I + H_2$$

$$B_2H_6 + HBr \rightarrow B_2H_5Br + H_2$$

$$B_2H_6 + HCl \rightarrow B_2H_5Cl + H_2$$

10) Action of halogens

Halogens also react with diborane to form the corresponding halodiboranes.

The reactivity decreases in the order: $Cl_2 > Br_2 > I_2$.

Thus Cl_2 reacts explosively at room temperature, Br2 reacts rapidly at 373 K while I_2 reacts slowly at still higher temperature.

$$B_2H_6 + Cl_2 \rightarrow B_2H_5Cl + HCl$$

Uses of Boron Hydrides

(1) Dibornae is used for preparing a number of boron hydrides such as $LIBH_4$, $NaBH_4$.

(2) It is used as a reducing agent in organic reactions.

(3) It is used as a fuel for supersonic rockets.

Structure of Diborane

1) In diborane, each boron atom has 3 Valence electrons for sharing. If we assume that each boron forms 3 covalent bonds with 3 hydrogen atoms, then there are no electrons left with boron atom for sharing with other boron atoms. Therefore the two boron atoms cannot be linked.

There are two types of hydrogen atom:

1) The four hydrogen atoms (two on left and two on right shown by thick lines) are called terminal hydrogens.
These four terminal hydrogens and the two boron atoms lie in the same plane while the remaining two hydrogen atoms-one lying above and one lying below this plane form bridges and hence are called which bridged hydrogen.
bana are two types of bonds in B_2H_6 molecules.
1) The four terminal B-H bonds are normal covalent bonds and hence are quite strong. Each bond is formed by sharing a pair of electron between B and H and hence are also called two centre electron pair bonds or two centre two electron bond (2c – 2e)
2) The two bridge bonds i.e. B — H — B are quite different from normal covalent bonds. Each bridge hydrogen is bonded to two boron atoms by a pair of of electrons. These are three centre electron pair bonds or three centre 2 electron bonds and hence are quiet weak.
Because of their semblance to a banana, these three centre electron pair bonds are also called banana Bond.

Structure of diborane on the basis of hybridization

The boron atom in the excited state is $1s^2\ 2s^1\ 2p_x^1\ 2p_y^1\ 2p_z^0$.
It undergoes sp^3 hybridisation
The two half filled hybrid orbitals of each boron atom overlap with the half filled orbitals of hydrogen atom forming normal covalent bonds whereas the third half filled hybrid orbital of one boron atom and the vacant hybrid orbital of the second boron atoms overlap simultaneously with the half filled orbitals of hydrogen atom The electron cloud contains only two electrons but spreads over 3 atoms. That is why this bond is called three centre electron pair bond.
Because of the shape of the electron cloud formed, it is called banana bond. Molecules like B2H, which do not have sufficient number of electrons to form normal covalent bonds are called electron deficient molecules.

Compounds of Boron

1. Borax/ Sodium Tetraborate ($Na_2B_4O_7.10H_2O$)

It is the most important compound of boron. It is a white crystalline solid. Borax dissolves in water to give an alkaline solution.

Preparation:

From Boric acid: Boric acid is neutralised with sodium carbonate and the resulting solution is cooled to get crystals of borax.

$$H_3BO_3 + Na_2CO_3 \rightarrow Na_2B_4O_7 + H_2O + CO_2$$

Properties:

i. It gets hydrolysed with water to form an alkaline solution

$$Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + H_3BO_3$$

ii. Borax bead test: On heating borax first swells up due to elimination of water molecules.

On further heating it melts to a liquid which then solidifies to a transparent glassy mass.

$$Na_2B_4O_7.10H_2O \rightarrow Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \rightarrow 2NaBO_2 + B_2O_3$$

iii. It is a useful primary standard for titration against acids.

$$Na_2[B_4O_5(OH)_4]\cdot 8H_2O + 2HCl \rightarrow 2NaCl + 4H_3BO_3 + 5H_2O$$

2. Diborane: B_2H_6

The simplest boron hydride known, is diborane. It is prepared by treating boron trifluoride with $LiAlH_4$ in diethyl ether.

Preparation

$$3LiAlH_4 + 4BCl_3 \rightarrow 3LiCl + 3AlCl_3 + B_2H_6$$

Properties

i. Stable at low temperature only, colourless and highly toxic.

ii. $B_2H_6 + 6H_2O \rightarrow 2H_3BO_3 + 6H_2$

iii. $B_2H_6 + 6Cl_2 \rightarrow 2BCl_3 + 6HCl$

iv. $B_3H_6 + 2Me_3N \rightarrow 2[Me_3N.BH_3]$

Uses of Boron and Aluminium and Their Compounds

Boron Compounds

Boron is a hard solid having high melting point low density and very low electrical conductivity. Some important boron compounds are:

1. **Boron fibers:** It is mixed with plastic to form a material which is lighter than aluminium but tougher and stiffer than steel hence it is used in body armour, missiles and aircrafts.
2. **Boron-10 (^{10}B) isotope:** Boron carbide rods or boron steel are used to control nuclear reactions as neutron absorbers.

   $_5B^{10} + {}_0n^1 \rightarrow {}_5B^{11}$
3. **Borax:** It is used in manufacture of enamels and glazes for pottery and tiles. It is also used in making optical glasses and also borosilicate glasses which is very

resistant to heat and shock. It is used as an antispectic.

4. **Boric acid:** It is used in glass industry, in food industry as preservative. It is also used as an antiseptic and eye wash under the name 'boric lotion'. It is also used in manufacture of enamels and glazes for pottery.
5. **Boron carbide:** Hardest boron compound.

Aluminium Compounds

Aluminium and its alloy are used in packing industry, utensil industry, aeroplane and transportation industry etc.

1. Alumina ($Al2O_3$):

a) Used in chromatography.

b) Used in making bauxite bricks which are used for lining furnaces.

2. Aluminium chloride ($AlCl_3$): Used in manufacture of dyes, drugs and perfumes and also in manufacture of gasoline. It is also used as catalyst in Friedel Craft reaction.

3. Potash Alum. [K_2SO_4-$Al_2(SO_4)_3$-24 H_2O]: Used in purification of water, leather tanning, as antiseptic and as a mordant.

Group 14 Elements: The Carbon Family

Group IV A contains six elements: carbon, silicon, germanium, tin, lead and ununquadium. The penultimate shell (prior to outermost) contains $1s^2$ -grouping in carbon, $2s^22p^6$ (8 electrons) in silicon and $(n-1)s^2(n-1)p^6(n-1)d^{10}$ (18 electrons) in other elements. This shows why carbon differs from silicon in some respects and these two differ from rest of the members of this group. General electronic configuration is ns^2np^2.

1. Atomic and Physical Properties

The important properties of carbon family are discussed below:

i. Atomic Radii: The atomic radii of group 14 elements are less than the corresponding elements of group 13. However, the atomic radii increases down the family.

ii. Ionisation Energies: The higher ionisation energies than group 13 are due to the higher nuclear charge and smaller size of atoms of group 14 elements. While moving down the group, the ionisation energies decreases till Sn.

C > Si > Ge > Sn < Pb

iii. Oxidation state and valency: The elements of group 14 show tetravalency by sharing four of its valence electrons. Therefore, they have oxidation state of +4. In addition, Ge, Sn and Pb also show +2 oxidation state.

iv. Catenation: Catenation is ability of like atoms to link with one another through covalent bonds. Tendency decreases from C to Pb. It is due to the decreasing M-M single bond energy. Thus, the tendency for catenation decreases as:

C > Si > Ge > Sn > Pb

v. Allotropy: All the elements of the carbon family with the exception of lead exhibit allotropy.

Carbon exists as two important allotropic forms diamond and graphite.

2. Chemical Properties

i. Reactivity towards air: All members of this group form monoxide of the general formula MO such as CO, SiO, SnO and PbO. All members of this group form dioxides of molecular formula MO_2 such as CO_2, SiO_2, GeO_2, SnO_2 and PbO_2.

ii. Reactivity towards water: In this family three members i.e., carbon, silicon and germanium are affected by water while lead is not affected by water due to formation of protective oxide film, but tin decomposes with steam into tin dioxide and hydrogen gas.

iii. Reactivity towards halogen: These elements form two types of hallides MX_2 and MX_4 Most of the MX_4 are covalent. SnF_4 and PbF_4 are ionic in nature.

Anomalous Behaviour of Carbon

Carbon shows anomalous behaviour due to its smaller size, higher electronegativity, higher ionization enthalpy and unavailability of d orbitals. Carbon atom forms double or triple bonds involving pπ-pπ bonding. Carbon has also the property to form closed chain compounds with O, S and N atoms as well as forming pπ-pπ multiple bonds with other elements particularly N, S and O. When we move down the group size increases and electronegativity decreases hence catenation tendency decreases. Order is:

$$C >> Si > Ge = S$$

Allotropes of Carbon

Carbon shows allotropism due to catenation and pл–pл bond formation. Carbon exists in two allotropic forms – crystalline and amorphous. The crystalline forms are diamond and graphite while the amorphous forms are coal, charcoal and lamp-black. The third form is fullerenes discovered by Kroto, Smalley and Curl.

Note: Tin has maximum number of allotropes.

Diamond

In diamond each carbon is joined to other four carbon tetrahedrally and carbon-carbon bond length is 1.54Å and bond angle is $109^{*}28'$ having sp3 hybridisation on each carbon. All four electrons in carbon are involved in bonding hence, it is bad conductor of electricity. Diamond is an excellent thermal conductor.

It is hardest natural substance known. It is transparent and has a specific gravity 3.52 and its refractive index is high (2.45). Difficult to break due to extented covalent bonding. Diamond is used for making cutters. Blades of diamond are used in eye surgery and as an abrasive for sharpening hard tools. Impure diamonds (black) are used in knives for cutting glass.

Graphite

Each carbon is sp^2 hybridised. It has layered structure. These layers are attracted by van der Waals force. Each carbon has one free electron in p-orbital, so it is a good conductor of electricity. All electrons get delocalized in one layer and form π-bond. Electron jumps from one orbital to another hence it is a good conductor of heat and electricity. In graphite carbon- carbon bond length is 141.5 pm and distance between adjacent graphite layer is 340 pm.

Graphite is used as a lubricant at high temperature. Oil gets burn or denatured at high temperature but graphite does not get denatured even at high temperature so, preferred over oil and grease.

Fullerene

It was made as a result of action of a laser beam or strong heating of a sample of graphite in presence of inert atmosphere. The sooty material mainly contains C_{60} with C_{70} (small amount). Most common fullerene is C_{60} called Buckminsterfullerene which has football-like structure. It contains 20 six-membered ring and 12 five-membered ring. It is used to make ball bearings.

Coal

It is the crude form of carbon. It has been formed in nature as a result of slow decomposition of vegetable matter under the influence of heat, pressure and limited

supply of air. The successive stages of transformation are peat, lignite, bituminous, steam coal and anthracite. Bituminous is hard stone, burns with smoky flame. The superior quality is anthracite which burns with non- smoky flame.

Uses of carbon

Graphite: In making lead pencils, electrodes of electric furnances, as a moderator in nuclear reactor, as a lubricant in machinery.

Charcoal: In removing offensive odour from air, in removing fused oil from crude spirit, in decolourising sugar syrup, in gas masks etc.

Carbon black: For making printing inks, black paints, Indian inks, boot polishes and ribbons of typewriters.

Coal: For the manufacture of coal gas, coal tar, coke, and synthetic petrol.

Allotropes of Carbon

1. Carbon Monoxide (CO)

Preparation: Carbon monoxide is majorly prepared by

$$2C+O_2 \rightarrow 2CO$$

Properties:

i. Burns with blue flame

$$2CO+O_2 \rightarrow 2CO_2$$

ii. $CO + Cl_2 \rightarrow COCl_2$ (Phosgene)

iii. $CO + 2H_2 \rightarrow CH_3OH$

iv. Many of the transition metals form metal carbonyls

$$Ni + 4CO \rightarrow Ni(CO)_4$$

2. Carbon Dioxide (CO_2)

Preparation:

Carbon dioxide is mostly prepared by decomposition of carbonates and bicarbonates:

(i). $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$

(ii). $CaCO_3 \rightarrow CaO + CO_2$

Properties:

Carbon dioxide is an acidic, colourless gas. The important properties are:

(i). $Zn + CO_2 \rightarrow ZnO + CO$

(ii). $2Mg + CO_2 \rightarrow 2MgO + C$

(iii). $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$

(iv). $Na_2CO_3 + H2O + CO_2 \rightarrow 2NaHCO_3$

Silicones

1) Silicones are synthetic organosilicon polymers containing repeated R_2SIO units held by Si-O-Si linkages.

These compounds have general formula $(R_2SiO)_n$ where R = methyl or aryl group.

Preparation

When methyl chloride reacts with silicon in the presence of copper as a catalyst at 570 K, various types of methyl substituted chlorosilanes of formula CH_3SiCl_3, $(CH_3)_2SiCl_2$, $(CH_3)_3SiCl$ along with small amount of $(CH_3)_4Si$ are formed.

Hydrolysis of dichlorodimethylsilanes followed by polymerization yields chain polymers.

Properties

(1) Silicones made up of short chain molecules are oily liquid, silicones with medium chains behave as viscous oils, jellies and greases, those with very long chains behave as rubbery elastomers and resins.

(2) They are chemically inert i.e. resistant to oxidation, thermal decomposition or to attack by organic reagents.

(3) Silicones being surrounded by non-polar alkyl group are water repelling in nature.

(4) They are heat resistant and have high dielectric strength i.e. have good electrical insulating properties.

Uses

1) For making water proof papers, wool, textile, wood by coating them with a thin film of silicone.

2) As sealants and electrical insulators.

3) As lubricant at high as well as low temperature since there is little change in their viscosities with temperature.

4) Being biocompatible, they are used in surgical and cosmetic implants.

Summary-

1. Allotropes: Those compounds which have different physical properties but similar chemical properties are called allotropes e.g., diamond and graphite.

2. Catenation: Tendency of carbon atom to link with itself to form long chain is called catenation.

3. Inert pair effect: Decrease in tendency of ns2 electron pair to participate in bond formation with increase in atomic number is called inert pair effect.

4. Silicones: It is organosilicon compound containing repeated R_2SiO units.

5. Silicates: Compounds in which anions are either discrete SiO_4^{-4} units or a number of such units combine together through corners.

6. Alums: All double sulphates having one monovalent basic radical and one trivalent basic radical.

7. Boranes: Hydrides of boron are called as boranes.

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